

Physics Week 9 Pilot Reproducibility Note

Pilot record. Use for protocol design, not final accuracy claims.

Reading guide. This is a reproducibility note, not a leaderboard. G0 has the highest exact agreement in this metrics file (91.3%). The historical G0 baseline is retained for context at 91.3% exact agreement. *Because this is a legacy pilot, these numbers guide protocol design and should not be presented as final cross-course evidence.*

What Is Being Reproduced

1. Freeze

Anonymous data snapshot and rubric are identified by hash.

2. Packetize

Model-facing work happens only inside prompt packets with `prompt.txt`.

3. Evaluate

Predictions are compared with the same metrics script and recorded here.

This record covers `physics / week9` on branch `codex/repro-experiment-framework`.

Reproducibility Anchors

Anchor	Value
Experiment	physics-week9-pilot
Git	codex/repro-experiment-framework @ 9e1c7c018268
Data snapshot	d134f9127adc...
Metrics	Data/physics/benchmark/metrics-all.json
Typst source	experiments/records/physics-week9-pilot/note.typ

Prompt Packet Registry

Condition	Packet SHA-256
G2-dev-r1	752bf779d5fb...
G2-test-r1	02d067965bfb...
G3-dev-r1	6738cb00a40d...
G3-test-r1	87103669cc7b...
T1-dev-r1	7763fc740353...
T1-test-r1	b63aadd3cf33...

Key Findings

- G0 is the highest exact-agreement condition in this file at 91.3%.
- The historical G0 baseline remains the top exact-agreement condition, so this pilot does not show a workflow improvement over G0.
- D2 has the highest severe-error rate (40.0%), which is the clearest operational risk signal in this pilot.
- DeepSeek transcript subset: D1 to D2 changed exact agreement from 78.3% to 79.2%; human-minus-automatic interval −10.0% to 15.8%.
- GPT transcript subset: G2 to G3 changed exact agreement from 87.5% to 78.3%; human-minus-automatic interval −11.7% to −6.7%.
- Interpret all findings as protocol guidance only; this pilot is not evidence that one workflow generalizes across courses.

Results At A Glance

Best exact

91.3%

G0

G0 exact

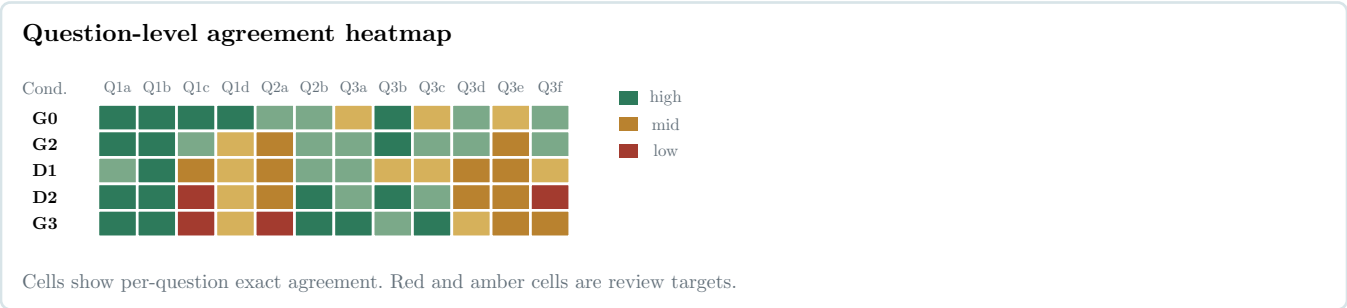
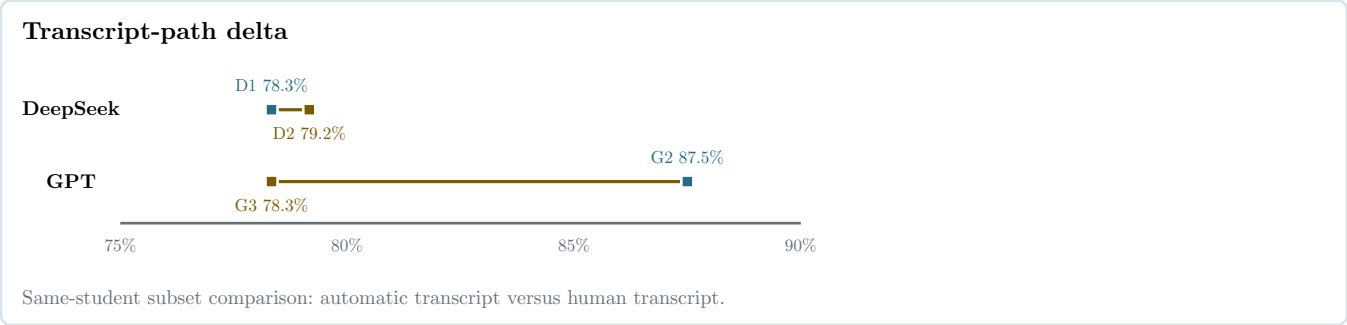
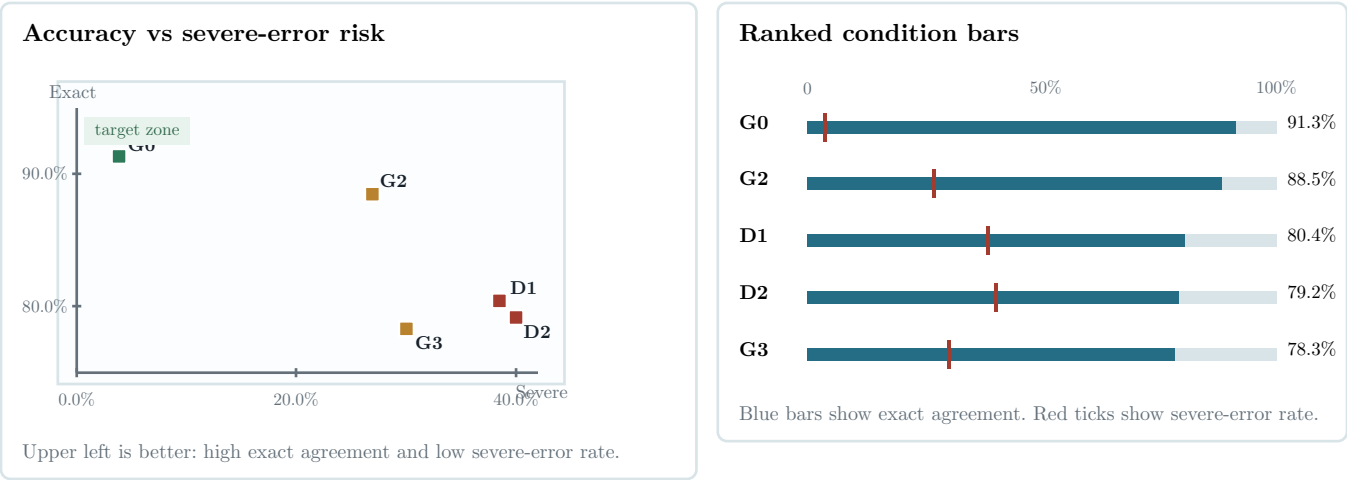
91.3%

historical baseline

Highest severe

40.0%

D2



Condition Details

Cond.	N	Population	Exact	Total MAE	Within 1	Severe	Bias
D1	26	full_split	80.4%	2.452	38.5%	38.5%	−0.169
D2	10	transcript_subset	79.2%	1.950	30.0%	40.0%	−0.154
G0	26	full_split	91.3%	0.413	88.5%	3.8%	+0.030
G2	26	full_split	88.5%	1.135	61.5%	26.9%	−0.027
G3	10	transcript_subset	78.3%	1.575	40.0%	30.0%	−0.119

Where Errors Concentrate

Condition	Lowest-agreement questions
D1	Q1c: 69.2%, Q2a: 69.2%, Q3d: 69.2%
D2	Q1c: 40.0%, Q3f: 50.0%, Q2a: 60.0%
G0	Q3e: 76.9%, Q3a: 80.8%, Q3c: 84.6%
G2	Q2a: 69.2%, Q3e: 73.1%, Q1d: 84.6%
G3	Q2a: 30.0%, Q1c: 40.0%, Q3e: 60.0%

Paired And Transcript Comparisons

Paired exact-agreement differences versus G0

Condition	Mean diff.	95% interval
D1	−10.9%	−18.3% to −4.2%
G2	−2.9%	−6.4% to 0.6%

Transcript subset comparisons

Model	Conditions	Auto exact	Human exact	Interval
DeepSeek	D1 -> D2	78.3%	79.2%	−10.0% to 15.8%
GPT	G2 -> G3	87.5%	78.3%	−11.7% to −6.7%

Reproduction Commands

```
python -m benchmark.core.cli audit-packet --packet <packet-path>
python -m benchmark.physics.cli validate --root Data/physics/benchmark
python -m benchmark.physics.cli evaluate --root Data/physics/benchmark --split dev
python -m benchmark.physics.cli evaluate --root Data/physics/benchmark --split test
python -m benchmark.physics.cli evaluate --root Data/physics/benchmark --split all
```

Limitations

- after-the-fact record for the legacy Physics Week 9 pilot
- pilot only; not a final accuracy conclusion
- private data is excluded from Git and must come from the private data snapshot
- reference scores are single-primary-rater, not adjudicated
- Physics Week 9 is a pilot and must not be generalized to other courses
- historical baselines and interactive model runs are not controlled reruns